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2027 CYCLE

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CUET UG

Chemistry

Alcohols, Phenols and Ethers

Official Syllabus Unit VII · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit VII)	Skills the questions test
Classification, naming, structures	Classification & nomenclature of alcohols, phenols & Ethers, Structures of functional groups (alcohol, phenols & ethers)	Sorting alcohols as 1°/2°/3°, mono/di/tri-hydric, spotting allylic/benzylic/vinylic positions, and writing -ol/alkoxy names.
Preparation	Methods of preparation of alcohols, Phenols and Ethers	Choosing acid hydration vs hydroboration (Markovnikov vs anti-Markovnikov), reductions, Grignard builds, the cumene/Dow/diazonium routes to phenol, and Williamson vs dehydration for ethers.
Physical properties	Physical properties of alcohols, phenols and Ethers (boiling point & solubility)	Ranking boiling points via hydrogen bonding (alcohol > ether > alkane) and explaining solubility fall with chain length.
Chemical properties	Chemical properties of alcohols involving cleavage of -C-OH bond & -CO-H bond, Chemical properties of phenols including oxidation & Reduction, Chemical properties of Ethers	Predicting Lucas-test speeds, dehydration products at 413 K vs 443 K, oxidation outcomes by alcohol class, phenol's acidity and ring reactions (Kolbe, Reimer-Tiemann), and ether cleavage by HI.
Commercial alcohols	Some Commercially important Alcohols like methanol & Ethanol	Recalling methanol's toxicity and CO route, ethanol's fermentation, azeotrope and denaturing.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit VII is the functional-group unit where reagent-choice logic decides nearly every answer.

Classification and Naming: Reading the Addresses

One hydroxyl group, many homes. When OH hangs on an sp^3 carbon, you have an **alcohol**; when it sits **directly on the ring**, you have a **phenol**; when oxygen bridges two carbons, you have an **ether**. The same O–H chemistry runs through the first two families, but the ring rewrites phenol's personality completely — this chapter is largely the story of that contrast.

- **Alcohols by the carbinol carbon: primary** (one carbon neighbour), **secondary** (two), **tertiary** (three) — the same 1°/2°/3° ladder that steered Unit VI's S_N questions, because converting an alcohol to a halide follows them too.
- **By how many OH:** monohydric (C_2H_5 OH), dihydric (ethylene glycol, CH_2 OH − CH_2 OH), trihydric (glycerol) — antifreeze and cosmetics own the latter two.
- **By position flavour: allylic** (OH next to a C=C on an sp^3 carbon), **benzylic** (next to the ring), **vinylic** (OH directly ON the C=C carbon — rare, unreactive, and excluded from the alcohol club in practice).
- **Names:** the -ol suffix with the lowest locant — propan-2-ol, ethane-1,2-diol, propane-1,2,3-triol. In rings, cyclohexanol; when OH is not the senior group, it becomes a **hydroxy** prefix.
- **Phenols:** the parent plus methylated cresols (o-, m-, p-cresol) and the dihydroxy trio — catechol (1,2), resorcinol (1,3), hydroquinone (1,4).
- **Ethers:** name the smaller side as an **alkoxy** prefix on the bigger chain — methoxyethane ($CH_3OCH_2CH_3$), ethoxybenzene. Symmetrical ones get the "di" form: diethyl ether.

Structures of the functional groups

- **Alcohol O:** both bonds from sp^3 oxygen — C–O–H angle near 108.9°, C–O length 142 pm.
- **Phenol O:** sp^2-flavoured (lone pair delocalised into the ring) — C–O shortens to 136 pm and the ring's electron density rises; every phenol specialty flows from this delocalisation.
- **Ether O:** sp^3 like the alcohol but two carbon friends; bent shape, polar C–O bonds, and **no O–H bond at all** — no self-hydrogen-bonding, and much lower boiling points.

Preparation: Building the O–H and C–O

Making alcohols

- **From alkenes — acid hydration (Markovnikov):** $CH_3 - CH = CH_2 + H_2O \xrightarrow{H^+} CH_3 - CH(OH) - CH_3$ — the OH lands on the **more substituted** carbon, via the more stable carbocation.

- **From alkenes — hydroboration-oxidation (anti-Markovnikov):** BH_3 adds boron to the **less** crowded carbon; $H_2\frac{O_2}{OH^-}$ then swaps B for OH — propene gives **propan-1-ol**. The clean way to beat Markovnikov.
- **By reduction:** aldehydes → 1° alcohols, ketones → 2° (reagents: $NaBH_4$, $LiAlH_4$); carboxylic acids and esters → 1° with $LiAlH_4$.
- **From Grignard reagents:** R–MgX + formaldehyde → 1° (one carbon longer); + other aldehydes → 2°; + ketones → 3°. Hydrolysis releases the alcohol. Chain-building on demand.

Making phenols

- **Dow process:** chlorobenzene + NaOH at 623 K, 300 atm, then acidification — brute force (Unit VI revisited).
- **From diazonium salts:** warm $ArN_2^+Cl^-$ with water → phenol + N_2 — the mild, classic lab route.
- **Cumene process (industrial champion):** benzene + propene → cumene; air oxidation → cumene hydroperoxide; dilute acid splits it into **phenol + acetone** — two valuable products from one reaction, which is why industry loves it.

Making ethers

- **Dehydration of alcohols (413 K, conc. H_2SO_4):** two alcohol molecules lose water between them — $2R$–OH → R–O–R + H_2 O. Limited to 1° alcohols (2°/3° prefer to eliminate), and gives symmetrical ethers only.
- **Williamson synthesis (the general route):** an alkyl halide + sodium alkoxide: R–X + $R'O^-Na^+$ → R–O–R' + NaX. Works for any ether **if the halide is primary** — with secondary/tertiary halides, the alkoxide (a base!) eliminates instead. To make an aryl alkyl ether, use the **sodium phenoxide** + alkyl halide: $C_6H_5O^-Na^+ + CH_3$ Br → $C_6H_5OCH_3$.

Physical Properties: The Hydrogen-Bond Tax

- **Boiling points:** alcohols tower over alkanes and ethers of similar mass because their O–H bonds network molecules together — ethanol (M = 46) boils at 351 K while dimethyl ether (also M = 46, but no O–H) boils at 248 K. Same mass, 100 K apart: that is the hydrogen-bond tax.
- **Among isomeric alcohols:** branching lowers boiling point (less surface contact) — the 1° > 2° > 3° order again.
- **Solubility:** methanol and ethanol mix with water in every proportion; solubility falls as the hydrophobic chain grows (octanol barely dissolves). Within isomers, branching **raises** solubility.
- **Phenol:** only moderately water-soluble — the ring dominates; its o-nitro cousin hydrogen-bonds **internally** (chelation), which is why o-nitrophenol is steam-volatile while the p-isomer, hydrogen-bonding **externally**, is not. A beloved separation question.
- **Ethers:** no O–H means no self-association — low boiling, but still **water-miscible in small members** (their bent oxygen accepts water's H-bonds).

Alcohols: Two Bonds, Two Attack Plans

The reactions of alcohols split by **which bond breaks**: the C–OH bond (substitutions, dehydration) or the O–H bond (acidity, esterification).

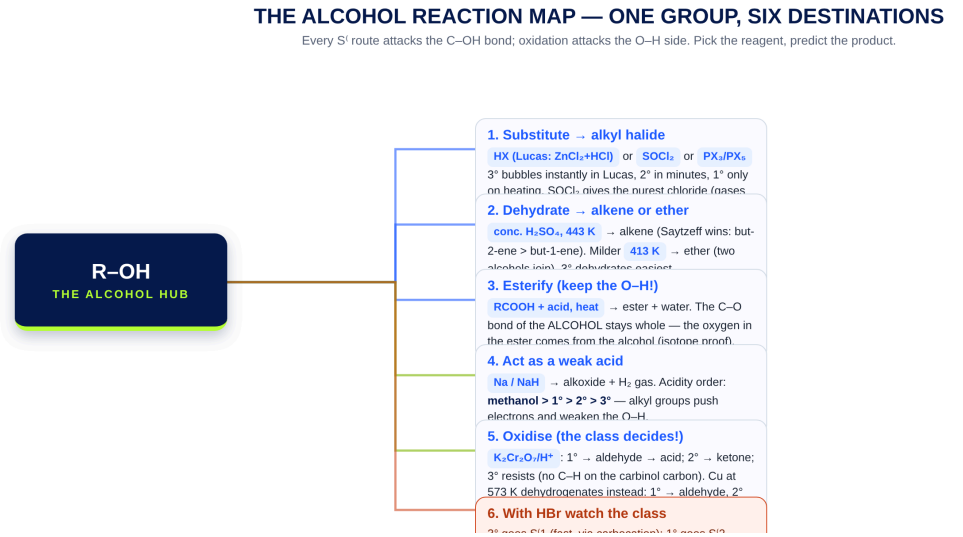


Figure 1: The alcohol reaction map: six destinations from one R–OH hub — halides, alkenes or ethers by dehydration, esters, alkoxides, and oxidation products chosen by alcohol class.

Cleaving the C–OH bond

- **With HX (the Lucas test):** R–OH + HX → R–X + H_2 O — Lucas reagent (HCl + $ZnCl_2$) reads the classes by turbidity: 3° clouds **instantly** (S_N1 via a stable carbocation), 2° in minutes, 1° only on heating (S_N2, slow). The order 3° > 2° > 1° is carbocation logic.
- **With phosphorus halides:** $3R$–OH + PCl_3 → $3R$–Cl + H_3PO_3; PBr_3 and PI_3 (from red P + Br_2/I_2) make the bromides and iodides.
- **With thionyl chloride:** R–OH + $SOCl_2$ → R–Cl + SO_2 → + HCl → — gaseous by-products escape, giving the purest alkyl chloride.
- **Dehydration — temperature chooses the product:** conc. H_2SO_4 at **413 K** joins two alcohols into an ether; at **443 K** each molecule loses water into an alkene. Ease of dehydration: 3° > 2° > 1° (carbocation again), and **Saytzeff's rule** picks the more substituted alkene.
- **Oxidation — the class fingerprint:** acidified $K_2Cr_2O_7$ turns 1° alcohols into aldehydes then (further) acids, and 2° into ketones; **3° refuses** — no C–H on the carbinol carbon to remove. With hot Cu at 573 K the route is **dehydrogenation**: 1° → aldehyde, 2° → ketone, 3° → alkene (dehydration wins instead).

Cleaving the O–H bond: acidity and esters

- **Acidity order:** methanol > 1° > 2° > 3° — alkyl groups push electrons (+I), weakening the O–H grip on its proton. All alcohols are **weaker acids than water**, so sodium metal on wet hands is wasted on water first; Na/NaH convert alcohols to **alkoxides** (RO^-), the workhorse bases of the Williamson synthesis.

- **Esterification:** R–OH + R'COOH $\xrightarrow{\text{acid}}$ R'COOR + H_2 O — and here is the isotope proof the exam enjoys: using 18 O-labelled alcohol, the heavy oxygen appears in the **ester**, not the water — so the alcohol's C–O bond never broke.

Phenols: The Ring Changes Everything

- **Acidity:** phenol (pKa ≈ 10) beats water and every alcohol, because the **phenoxide ion spreads its negative charge into the ring** by resonance — the charge never concentrates. Electron-pulling nitro groups stretch the difference further (picric acid, 2,4,6-trinitrophenol, is nearly a strong acid); alkyl groups shrink it.
- **Reactions of the O–H:** with Na → sodium phenoxide + H_2 (like alcohols); with NaOH → phenoxide (unlike alcohols — evidence of the higher acidity); **not** with $NaHCO_3$ (that distinguishes phenol from carboxylic acids, which do fizz).
- **Ring reactions — no catalyst, no heat:** the activated ring over-delivers. Bromine water **instantly** gives 2,4,6-tribromophenol as a white precipitate (no $FeBr_3$ needed); dilute HNO_3 at low temperature gives o- and p-nitrophenol.
- **Kolbe reaction:** sodium phenoxide + CO_2 under pressure, then acid → **salicylic acid** — the feedstock of aspirin.
- **Reimer-Tiemann reaction:** phenol + $CHCl_3$ / NaOH, then acid → **salicylaldehyde** (an aldehyde lands ortho to OH).
- **Oxidation and reduction:** chromic acid oxidises phenol to **quinone** (a yellow ring-diketone); catalytic hydrogenation forces the ring to cyclohexanol.
- **The classic test:** neutral $FeCl_3$ gives a **violet complex** — a one-tube identification of phenols.

Ethers: The Quiet Family

- **Inertness:** no O–H, no carbonyl, no leaving group — ethers dissolve almost anything and react with almost nothing, which is exactly why **dry ether** hosts Grignard reagents (the oxygen donates lone pairs to magnesium).
- **Cleavage by HI (the one reaction they own):** R–O–R' + HI → R–I + R'–OH at heat. One equivalent stops at alcohol + iodide; **excess** HI takes the alcohol on to a second iodide (2R–I). Mechanism splits by class: 1° alkyl goes S_N2 (I^- attacks the less hindered side), 3°/benzylic go S_N1 (carbocation first). With **aryl** alkyl ethers, the aryl–O bond refuses to break — you always get the alkyl iodide + phenol.
- **Electrophilic substitution:** the alkoxy group is a strong activator, o,p-directing — anisole nitrates gently to o- and p-nitroanisole.

Methanol and Ethanol: The Commercial Duo

- **Methanol ("wood spirit"):** made industrially from CO + $2H_2$ over a ZnO–"Cr"*2 "O"*3 catalyst. **Deadly poison** — metabolised to formaldehyde and formic acid; it blinds and kills. Never tasted, never sipped.
- **Ethanol:** the fermentation product — $C_6H_{12}O_6$ $\xrightarrow{\text{zymase}}$ $2C_2H_5$ OH + $2CO_2$. Distillation stalls at **95.6%** (a water-ethanol azeotrope); "absolute" alcohol needs drying (CuO, or molecular

sieves). **Denatured** ethanol carries methanol (or other additives) to make it undrinkable — industrial use without beverage tax.

- **Everyday anchor:** ethanol is the solvent of tinctures and perfumes; methanol the feedstock of formaldehyde.

Key Reactions & Rules: One-Stop Revision

ROUTES, ORDERS, NAMED REACTIONS

Rule / route	Symbols in plain words	When to reach for it
Alkene + $H_2\frac{O}{H^+}$	Markovnikov: OH to the richer carbon	Alcohol from alkene (default)
Hydroboration: BH_3 then $H_2\frac{O_2}{OH^-}$	anti-Markovnikov OH	Making 1° alcohols from alkenes
R MgX + HCHO / RCHO / RCOR'	1° / 2° / 3° alcohols after hydrolysis	Chain-building by Grignard
$LiAlH_4$, $NaBH_4$	reduce carbonyls to alcohols	Aldehyde → 1°, ketone → 2°
Cumene process	phenol + acetone from benzene + propene + air	Industrial phenol questions
Williamson: $R–X + R'ONa$	primary halides only; phenoxide for aryl ethers	Any ether synthesis
Dehydration: 413 K vs 443 K	ether vs alkene (Saytzeff)	Temperature-choice questions
Lucas test: HCl + $ZnCl_2$	3° instant, 2° minutes, 1° heat	Classifying alcohols fast
$SOCl_2$	purest chloride — gases escape	Clean R–Cl preparation
Oxidation by $K_2Cr_2O_7$	1° → aldehyde → acid; 2° → ketone; 3° resists	Class-fingerprint questions
Cu, 573 K	dehydrogenation: 1° → aldehyde, 2° → ketone, 3° → alkene	The Cu-vs-oxidant contrast
Acidity: PhOH > H_2 O > 1° > 2° > 3°; within: MeOH > 1° > 2° > 3°	phenoxide is resonance-stabilised	pKa ordering questions
Kolbe: phenoxide + CO_2	salicylic acid	Aspirin synthesis route

Rule / route	Symbols in plain words	When to reach for it
Reimer-Tiemann: $\frac{CHCl_3}{NaOH}$	salicylaldehyde	Formylation of phenol
$FeCl_3$ test	violet complex with phenols	Spot-test identification
Ethers + excess HI	1 eq: alcohol + iodide; excess: two iodides; aryl C–O never breaks	Cleavage-product predictions
Ethanol 95.6% azeotrope	absolute alcohol needs drying; denatured = + methanol	Commercial alcohol facts

Exceptions & Traps

THE TRAPS THIS CHAPTER SETS — READ BEFORE THE MOCK

The rule you assume	The exception / twist	The trap in the question
More alkyl groups = more acidic (as in carbocation stability)	Reversed for O–H: 3° alcohol is the **weakest** acid	Acidity ordering 1°/2°/3° — the +I push weakens O–H
Phenol and alcohol behave alike	Phenol dissolves in NaOH (alcohol does not) but not in $NaHCO_3$ (acids do)	The three-way separation: alcohol / phenol / acid
Hydration always gives Markovnikov	Hydroboration-oxidation is anti-Markovnikov	Propene → propan-1-ol vs propan-2-ol
Williamson works with any halide	2°/3° halides eliminate (alkoxide = base); use the other side instead	Making methyl tert-butyl ether: tert-butoxide + methyl bromide, never the reverse
Dehydration gives the alkene at any temperature	413 K gives ether; 443 K gives alkene	Product-vs-temperature MCQs
3° alcohols oxidise easily	They refuse — no carbinol C–H	"Which alcohol resists acidified dichromate" — the 3° one
Esterification breaks the alcohol's C–O	It survives — the isotope 18 O lands in the ester	The labelled-oxygen tracing question
Ethers split symmetrically with HI	Splits depend on class; aryl–O bond NEVER breaks	Anisole + HI always yields phenol + CH_3I

The Whole Chapter in One Picture

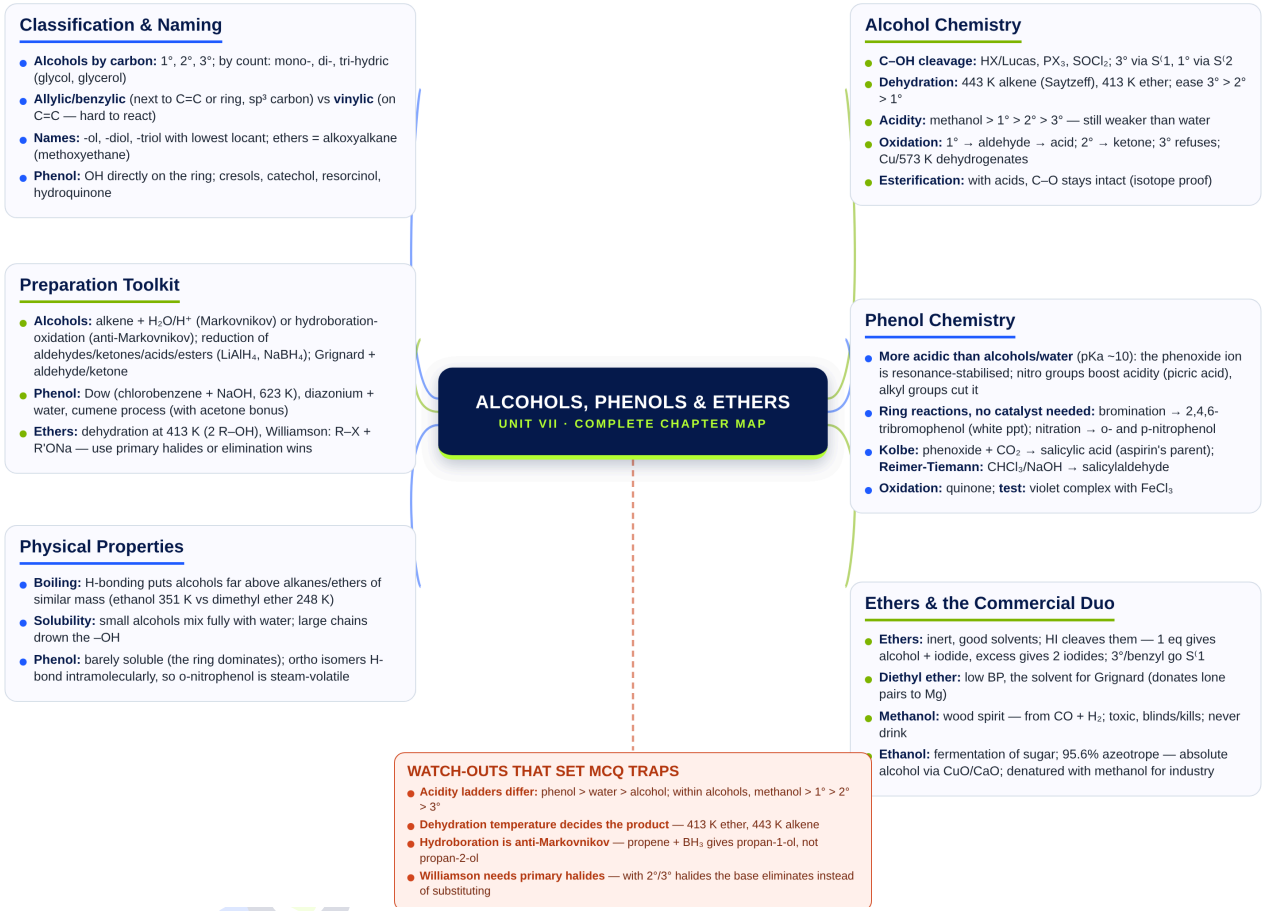


Figure 2: Chapter mindmap: classification and naming, the three preparation toolkits, hydrogen-bonded boiling points, the two attack plans of alcohols, phenol's ring chemistry, ether cleavage, and the commercial duo — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and takes nothing away for a wrong one.

Q1. Propene reacts with water in presence of dilute acid. The main product is:

(A) propan-1-ol
(B) propan-2-ol
(C) propane-1,2-diol
(D) propyl hydrogensulphate only

Q2. Which reagent converts propene to propan-1-ol (anti-Markovnikov)?

(A) $H_2\frac{O}{H^+}$
(B) BH_3 then $H_2\frac{O_2}{OH^-}$
(C) $LiAlH_4$
(D) $SOCl_2$

Q3. With Lucas reagent, the alcohol that clouds instantly is:

(A) ethanol **(B)** propan-1-ol
(C) propan-2-ol **(D)** 2-methylpropan-2-ol

Q4. The increasing order of acidity among alcohols is:

(A) $3° < 2° < 1° <$ methanol **(B)** methanol $< 1° < 2° < 3°$
(C) $1° < 2° < 3° <$ methanol **(D)** all equal

Q5. Heating ethanol with conc. H_2SO_4 at 413 K gives mainly:

(A) ethene **(B)** ethoxyethane
(C) acetaldehyde **(D)** ethyl hydrogensulphate

Q6. Phenol is more acidic than ethanol because:

(A) phenol has more carbon atoms **(B)** the phenoxide ion is stabilised by resonance
(C) phenol's O–H bond is longer **(D)** ethanol cannot hydrogen bond

Q7. Phenol + bromine water at room temperature gives:

(A) bromobenzene **(B)** 2,4,6-tribromophenol (white ppt)
(C) phenyl bromide + HBr only **(D)** no reaction without $FeBr_3$

Q8. The Kolbe reaction converts sodium phenoxide into salicylic acid using:

(A) CO_2, then acid **(B)** $\frac{CHCl_3}{NaOH}$
(C) CH_3 Cl **(D)** HNO_3

Q9. Anisole ($C_6H_5OCH_3$) with excess HI gives:

(A) CH_3I + phenol **(B)** methanol + iodobenzene
(C) two methyl iodides **(D)** no reaction

Q10. Absolute (100%) ethanol cannot be got by simple distillation because:

(A) ethanol decomposes **(B)** a 95.6% azeotrope forms
(C) ethanol is too volatile **(D)** water is denser

Detailed Solutions

Sol. Q1. Acid hydration follows Markovnikov: the OH lands on the carbon holding more alkyl groups — the secondary carbon. Propan-2-ol results; propan-1-ol needs hydroboration instead.

Answer: (B)

Sol. Q2. Hydroboration-oxidation is the anti-Markovnikov route: boron rides to the less crowded end, oxidation swaps it for OH. Propene → propan-1-ol. Acid water would have given propan-2-ol.

Answer: (B)

Sol. Q3. Instant turbidity means a fast S_N1 — a stable tertiary carbocation. 2-methyl-propan-2-ol is the tertiary member; the 1° and 2° options need minutes or heat.

Answer: (D)

Sol. Q4. Alkyl groups push electrons (+I) and weaken the O–H grip on H. So acidity **rises** as substitution falls: 3° < 2° < 1° < methanol — and every one of them is weaker than water.

Answer: (A)

Sol. Q5. Temperature decides: 413 K joins two ethanol molecules into ethoxyethane (ether); 443 K would dehydrate to ethene. The classic two-temperature question.

Answer: (B)

Sol. Q6. When phenol loses H^+, the negative charge spreads into the ring by resonance — the phenoxide ion never concentrates its charge. Ethoxide has no such escape. Resonance stabilisation = stronger acid.

Answer: (B)

Sol. Q7. The ring is so activated that bromine water reacts without any catalyst, instantly, at all three o/p positions: 2,4,6-tribromophenol, a white precipitate. $FeBr_3$ is for the **unactivated** benzene.

Answer: (B)

Sol. Q8. Kolbe: sodium phenoxide + CO_2 under pressure, then acidification → salicylic acid (OH and COOH neighbours). Reimer-Tiemann with $CHCl_3$/NaOH gives salicylaldehyde instead.

Answer: (A)

Sol. Q9. HI cleaves the alkyl–O bond only — the aryl C–O bond never breaks (resonance double-bond character). So anisole yields methyl iodide + phenol; excess HI leaves the phenol untouched.

Answer: (A)

Sol. Q10. Ethanol–water forms a minimum-boiling azeotrope at 95.6% ethanol — vapour and liquid share the composition, so distillation cannot cross it. Chemical drying (or molecular sieves) finishes the job.

Answer: (B)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option A	Assuming OH takes the simpler end	Markovnikov: the **richer** carbon gets the OH — only hydroboration reverses this
Q4 option B	Reusing carbocation-stability logic	Carbocations love alkyl groups; O–H acidity hates them — two opposite ladders
Q5 option A	Ignoring the temperature dial	413 K ether, 443 K alkene — the question always states one
Q7 option D	Requiring the catalyst out of habit	Phenol's ring needs NO catalyst — it reacts with bromine water instantly
Q9 option B	Cleaving the aryl bond	Resonance makes aryl–O unbreakable; the alkyl side always leaves as the iodide

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